

Nandini Patel

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SUMMARY

Software Engineer with 1+ years of expertise in road-to-cloud data pipeline engineering, ML model integration, & IoT edge system design, wielding Python, TypeScript, PostgreSQL, AWS, & OpenCV to ship high-throughput vehicle identification infrastructure that sustains 99.95% accuracy while processing millions of high-resolution images daily at scale

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, SQL, Bash

Frameworks & Tools: FastAPI, NestJS, Node.js, Next.js, React, Tailwind CSS, OpenCV

Databases: PostgreSQL, MongoDB, Redis, MySQL

Cloud & Infrastructure: AWS (EC2, S3), Docker, Linux, GitHub Actions, CI/CD Pipelines, IoT Edge Devices

Specialties: Computer Vision, ML Model Integration, Image Pipeline Engineering, Payment Processing, Sentry

EXPERIENCE

Software Engineer Intern (Data Pipeline & IoT Edge Systems)

Aug 2025 – Dec 2025

Electracast Media

- Architected road-to-cloud ingestion pipelines in Python & FastAPI, processing 2M+ sensor images on 4 AWS nodes.
- Spearheaded IoT edge data capture modules in Python & C++, achieving sub-80ms packet assembly on 6 sensors.
- Engineered PostgreSQL schemas & Redis caching layers, sustaining 99.95% accuracy across 3 live production nodes.
- Integrated OpenCV detection into image intake pipelines, lifting vehicle ID throughput by 4x per deployment run.
- Containerized edge-to-cloud services via Docker & Linux daemons, slashing release cycles from 3 days to 5 hours.
- Profiled distributed ingestion bottlenecks via Python tracing, resolving 8x critical throughput blockers in AWS.

Software Engineer (ML Integration & Computer Vision)

Feb 2024 – Jun 2024

Arcelor Mittal Nipon Steel

- Collaborated with ML teams to embed CV models into toll workflows, lifting overall vehicle match accuracy by 3x.
- Designed OpenCV & Python preprocessing for high-res frames, sustaining 98%+ image accuracy for 5M images/day.
- Optimized NestJS microservice APIs for ML inference routing, slashing model response latency from 420ms to 90ms.
- Wired MongoDB stores to CV pipelines, indexing 900K+ annotated vehicle records across 4 live production systems.
- Implemented canary rollouts for retrained CV models via Docker & GitHub Actions, maintaining 99.9% uptime daily.
- Validated model outputs via automated Python eval suites, diagnosing 11 regressions before each production push.

Software Engineer (Backend APIs, Payments & Cloud Infrastructure)

Jan 2023 – Jan 2024

PixML Solutions

- Developed NestJS & Node.js REST APIs for toll payment processing, serving 3 high-volume production lane systems.
- Provisioned AWS EC2 & S3 for distributed image storage, scaling capacity to 6M+ daily high-res asset transfers.
- Orchestrated PostgreSQL query optimization & connection pooling, reducing p95 read latency from 310ms to 55ms.
- Authored TypeScript validation modules enforcing 99.9% toll charge precision across all 5 billing integrations.
- Deployed Redis pub-sub event queues for real-time safety infraction alerts, handling 40K+ live events each hour.
- Managed Linux server provisioning & cron-based monitoring, sustaining 99.8% system uptime across 4 environments.

PROJECTS

Road-to-Cloud Image Ingestion Pipeline | Python, FastAPI, OpenCV, PostgreSQL, Redis, Docker, AWS

- Constructed an end-to-end image ingestion pipeline in Python & FastAPI, processing 1.2M+ frames on 3 AWS nodes.
- Integrated OpenCV vehicle detection into intake pipeline, achieving 97%+ classification accuracy on test frames.
- Deployed PostgreSQL & Redis layers via Docker on AWS EC2, sustaining sub-100ms responses under peak loads.
- Automated CI/CD via GitHub Actions, collapsing environment provisioning from 2 days to under ~4 hours per cycle.

ML Vehicle Classifier with NestJS API Layer | Python, TypeScript, NestJS, OpenCV, MongoDB, AWS, Docker

- Built a Python & OpenCV image preprocessing engine, normalizing 500K+ high-res vehicle frames for ML ingestion.
- Exposed NestJS & TypeScript REST APIs for realtime ML inference, serving classification results under 120ms avg.
- Stored annotated vehicle metadata in MongoDB, indexing 750K+ records to 4 downstream analytics dashboards.
- Deployed the full system stack to AWS via Docker, achieving 99.8% availability across 4+ production week cycles.

EDUCATION

University of Houston

Houston, TX

Master of Science in Computer Science (GPA: 3.8/4.0)

Aug 2024 – Dec 2025

Gujarat Technological University

Gujarat, India

Bachelor of Engineering in Computer Engineering (GPA: 3.9/4.0)

Aug 2020 – May 2024